

Exercises (with Solutions) · Chapter 3

1. Problem

Consider the matrix

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}.$$

- Determine A^T . Is A symmetric?
- Is A idempotent?
- Compute the determinant and the rank. Is A nonsingular?
- Compute the inverse.
- Compute the trace.

Solution

- The transpose of A is

$$A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}.$$

No, matrix A is not symmetric, since $A \neq A^T$.

- No, matrix A is not idempotent, since

$$AA = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = \begin{pmatrix} 7 & 10 \\ 15 & 22 \end{pmatrix} \neq A.$$

- Remember: $\det \begin{pmatrix} a & b \\ c & d \end{pmatrix} = ad - bc$. The determinant of A is

$$\det(A) = \det \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} = 1 \cdot 4 - 2 \cdot 3 = 4 - 6 = -2.$$

Since $\det(A) \neq 0$, matrix A

- has full rank and
- is nonsingular.

- Since A has full rank (and is nonsingular), the inverse of A exists and is given by

$$A^{-1} = \frac{1}{\det(A)} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = -\frac{1}{2} \begin{pmatrix} 4 & -2 \\ -3 & 1 \end{pmatrix} = \begin{pmatrix} -2 & 1 \\ 3/2 & -1/2 \end{pmatrix}.$$

- The trace of a matrix is the sum of the diagonal elements. For matrix A we have

$$\text{tr}(A) = 1 + 4 = 5.$$

2. Problem

- Let $AB = C$, where

$$A = \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix}.$$

Compute B .

- Let δ and γ be $c \times 1$ vectors, X be a $d \times c$ matrix, and Y be a $c \times d$ matrix. Determine the orders of

- XY ,
- YX ,
- $\gamma^\top \gamma$,
- $\gamma \gamma^\top$, and
- $\delta^\top YX \gamma$.

Under which conditions do the expressions Y^{-1} and $\delta^\top YX + \gamma^\top \gamma$ exist?

(c) Compute $\text{tr}(\lambda R^\top R)$ for $\lambda \in \mathbb{R}$ and

$$R = \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}.$$

Solution

(a) To solve the equation

$$AB = C$$

for B , we multiply both sides by A^{-1} from the left:

$$\begin{aligned} A^{-1}AB &= A^{-1}C \\ B &= A^{-1}C, \end{aligned}$$

where A^{-1} exists, because $\det(A) = \det \begin{pmatrix} 1 & 0 \\ 2 & 2 \end{pmatrix} = 1 \cdot 2 - 2 \cdot 0 = 2 \neq 0$. We have

$$A^{-1} = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ -2 & 1 \end{pmatrix},$$

Thus

$$B = A^{-1}C = \frac{1}{2} \begin{pmatrix} 2 & 0 \\ -2 & 1 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & 5 \end{pmatrix} = \begin{pmatrix} 1 & 3 \\ 0 & -1/2 \end{pmatrix}.$$

(b) The orders of the expressions are as follows:

- $\underset{(d \times c)(c \times d)}{X \ Y}$ has order $(d \times d)$.
- $\underset{(c \times d)(d \times c)}{Y \ X}$ has order $(c \times c)$.
- $\underset{(1 \times c)(c \times 1)}{\gamma^\top \ \gamma}$ has order (1×1) ; is a scalar.
- $\underset{(c \times 1)(1 \times c)}{\gamma \ \gamma^\top}$ is a matrix of order $(c \times c)$, and
- $\underset{(1 \times c)(c \times c)(c \times 1)}{\delta^\top YX \ \gamma}$ has order (1×1) ; i.e. is a scalar.

Y^{-1} exists if Y is a square, nonsingular matrix.

$\delta^\top YX + \gamma^\top \gamma$ exists if the terms have compatible orders/dimensions:

$$\underbrace{\underset{(1 \times c)}{\delta^\top} \underset{(c \times d)}{Y} \underset{(d \times c)}{X}}_{(1 \times c)} + \underbrace{\underset{(1 \times c)}{\gamma^\top} \underset{(c \times 1)}{\gamma}}_{(1 \times 1)}$$

Since $\gamma^\top \gamma$ is a scalar, it must hold that $c = 1$.

(c)

$$\begin{aligned}
\lambda R^\top R &= \lambda \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix}^\top \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix} \\
&= \lambda \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix} \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix} \\
&= \lambda \begin{pmatrix} \frac{1}{4} \cdot \frac{1}{4} + \frac{\sqrt{3}}{4} \cdot \frac{\sqrt{3}}{4} & \frac{1}{4} \cdot \frac{\sqrt{3}}{4} + \frac{\sqrt{3}}{4} \cdot \frac{3}{4} \\ \frac{\sqrt{3}}{4} \cdot \frac{1}{4} + \frac{3}{4} \cdot \frac{\sqrt{3}}{4} & \frac{\sqrt{3}}{4} \cdot \frac{\sqrt{3}}{4} + \frac{3}{4} \cdot \frac{3}{4} \end{pmatrix} \\
&= \lambda \begin{pmatrix} \frac{1}{4} & \frac{\sqrt{3}}{4} \\ \frac{\sqrt{3}}{4} & \frac{3}{4} \end{pmatrix} \\
&= \begin{pmatrix} \frac{\lambda}{4} & \frac{\sqrt{3}\lambda}{4} \\ \frac{\sqrt{3}\lambda}{4} & \frac{3\lambda}{4} \end{pmatrix}
\end{aligned}$$

Thus

$$\text{tr}(\lambda R^\top R) = \frac{\lambda}{4} + \frac{3\lambda}{4} = \lambda.$$

3. ProblemLet A be a nonsingular $(m \times m)$ matrix. Simplify the expression as far as possible:

$$\left(\frac{1}{\sqrt{2}} A^{-1} \left(\frac{1}{\sqrt{2}} (A^\top)^\top + \frac{\sqrt{2}}{2} A \right) \right)^{-1}.$$

Solution

$$\begin{aligned}
\left(\frac{1}{\sqrt{2}} A^{-1} \left(\frac{1}{\sqrt{2}} (A^\top)^\top + \frac{\sqrt{2}}{2} A \right) \right)^{-1} &= \left(\frac{1}{\sqrt{2}} A^{-1} \left(\frac{1}{\sqrt{2}} A + \frac{1}{\sqrt{2}} A \right) \right)^{-1} && \left| \text{Using } (A^\top)^\top = A \text{ and } \frac{\sqrt{2}}{2} = \frac{1}{\sqrt{2}} \right. \\
&= \left(\underbrace{\frac{1}{\sqrt{2}} A^{-1}}_{=\sqrt{2}^{-1} A^{-1}} \underbrace{\left(\frac{1}{\sqrt{2}} A + \frac{1}{\sqrt{2}} A \right)}_{=\sqrt{2} A} \right)^{-1} \\
&= \left(\sqrt{2}^{-1} A^{-1} \sqrt{2} A \right)^{-1} \\
&= \left(1 \cdot I_m \right)^{-1} = I_m^{-1} = I_m
\end{aligned}$$

4. ProblemConsider the $n \times (p+1)$ matrix X with $\text{rank}(X) = p+1$. Moreover, let $P_X = X(X^\top X)^{-1}X^\top$, and let $M_X = I_n - P_X$

(a) Determine the order of the following matrices:

- I_n
- $X^\top X$
- P_X
- M_X

(b) Which matrices from (a) are symmetric?

(c) Which matrices from (a) are idempotent?

(d) Compute the trace of I_n and P_X .

Solution

(a) Matrix orders/dimensions:

- I_n is $n \times n$
- $X^\top X$ is $(p+1) \times (p+1)$
- P_X is $n \times n$
- M_X is $n \times n$

(b) Checking symmetry for each matrix:

- $I_n^\top = I_n$, thus I_n is symmetric.
- $(X^\top X)^\top = X^\top X^\top{}^\top = X^\top X$, thus $X^\top X$ is symmetric.
- Check for P_X :

$$\begin{aligned} P_X^\top &= (X(X^\top X)^{-1} X^\top)^\top \\ &= ((X(X^\top X)^{-1}) (X^\top))^\top \\ &= X^\top{}^\top (X(X^\top X)^{-1})^\top \\ &= X ((X^\top X)^{-1})^\top X^\top \\ &= X ((X^\top X)^\top)^{-1} X^\top \\ &= X (X^\top X)^{-1} X^\top = P_X \end{aligned}$$

Thus P_X is symmetric.

- Check for M_X :

$$\begin{aligned} M_X^\top &= (I_n - P_X)^\top \\ &= I_n^\top - P_X^\top \\ &= I_n - P_X = M_X \end{aligned}$$

Thus M_X is symmetric.

(c) Checking which matrices are idempotent:

- $I_n I_n = I_n$, thus I_n is idempotent.
- Generally, $(X^\top X)(X^\top X) \neq (X^\top X)$. Example:

$$(X^\top X)(X^\top X) = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix} = \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix}$$

Thus, $X^\top X$ is generally not idempotent.

- Check for P_X :

$$\begin{aligned} P_X P_X &= X(X^\top X)^{-1} \underbrace{X^\top X(X^\top X)^{-1}}_{I_{(p+1)}} X^\top \\ &= X(X^\top X)^{-1} I_{(p+1)} X^\top \\ &= X(X^\top X)^{-1} X^\top = P_X \end{aligned}$$

Thus P_X is idempotent.

- Check for M_X :

$$\begin{aligned}
 M_X M_X &= (I_n - P_X)(I_n - P_X) \\
 &= I_n I_n - I_n P_X - P_X I_n + P_X P_X \\
 &= I_n - P_X - P_X + P_X P_X \\
 &= I_n - P_X - P_X + P_X \\
 &= I_n - P_X = M_X
 \end{aligned}$$

Thus M_X is idempotent.

(d) Computing the trace of I_n and P_X :

$$\begin{aligned}
 \bullet \operatorname{tr}(I_n) &= \operatorname{tr} \begin{pmatrix} 1 & 0 & \dots & 0 \\ 0 & 1 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & 1 \end{pmatrix} = n \\
 \bullet \operatorname{tr}(P_X) &= \operatorname{tr} \underbrace{\left(X(X^\top X)^{-1} X^\top \right)}_{\text{trace trick}} = \operatorname{tr} \left((X^\top X)^{-1} X^\top X \right) = \operatorname{tr}(I_{(p+1)}) = p+1
 \end{aligned}$$

5. Problem

Let X be a $n \times (p+1)$ matrix. Show that $X^\top X$ is positive semi-definite. Under which condition is $X^\top X$ positive definite?

Solution

- For positive semi-definiteness, we need to check whether

$$c^\top X^\top X c \geq 0 \quad \text{for all vectors } c \in \mathbb{R}^{p+1}.$$

The $((p+1) \times (p+1))$ matrix $X^\top X$ is positive semi-definite, because for any vector $c \in \mathbb{R}^{p+1}$ we have

$$\begin{aligned}
 \underbrace{c^\top}_{(1 \times n)} \underbrace{X^\top}_{(n \times 1)} \underbrace{X c}_{(n \times 1)} &= \overbrace{(Xc)^\top}^{=:y^\top} \overbrace{(Xc)}^{=:y} \\
 &= y^\top y \\
 &= \sum_{i=1}^n y_i^2 \geq 0.
 \end{aligned}$$

- The $((p+1) \times (p+1))$ matrix $X^\top X$ is positive definite, if

$$c^\top X^\top X c > 0 \quad \text{for all nonzero vectors } c \in \mathbb{R}^{p+1}.$$

That is, the $((p+1) \times (p+1))$ matrix $X^\top X$ is **not** positive definite, if there exists at least one nonzero $((p+1) \times 1)$ vector $c \neq 0$ such that

$$c^\top X^\top X c = (Xc)^\top (Xc) = \underbrace{0}_{(1 \times 1)}.$$

The product $(Xc)^\top (Xc)$ is zero if one of the factors is zero; i.e. if

$$\begin{aligned}
 & \begin{matrix} Xc & = & 0 \\ (n \times 1) & & (n \times 1) \end{matrix} \\
 & \underbrace{\begin{pmatrix} x_{11} & x_{12} & \dots & x_{1(p+1)} \\ \vdots & \vdots & & \vdots \\ x_{n1} & x_{n2} & \dots & x_{n(p+1)} \end{pmatrix}}_{(n \times (p+1))} \underbrace{\begin{pmatrix} c \\ \vdots \\ c \end{pmatrix}}_{((p+1) \times 1)} = \underbrace{0}_{(n \times 1)} \\
 & \underbrace{\begin{pmatrix} x_1 & x_2 & \dots & x_{(p+1)} \end{pmatrix}}_{(n \times (p+1))} \underbrace{\begin{pmatrix} c \\ \vdots \\ c \end{pmatrix}}_{((p+1) \times 1)} = \underbrace{0}_{(n \times 1)} \\
 & \underbrace{x_1}_{(n \times 1)} \underbrace{c_1}_{(1 \times 1)} + \underbrace{x_2}_{(n \times 1)} \underbrace{c_2}_{(1 \times 1)} + \dots + \underbrace{x_{(p+1)}}_{(n \times 1)} \underbrace{c_{(p+1)}}_{(1 \times 1)} = \underbrace{0}_{(n \times 1)}
 \end{aligned}$$

for a nonzero $((p+1) \times 1)$ vector $c \neq 0$.

Thus, $X^\top X$ is positive definite, if the $(n \times (p+1))$ matrix X has full column rank, i.e. if

$$\text{rank}(X) = p+1.$$

6. Problem

Let $y \in \mathbb{R}^n$, X be a $n \times (p+1)$ matrix, $b \in \mathbb{R}^{(p+1)}$ and let

$$f(b) = (y - Xb)^\top (y - Xb).$$

Compute the following derivatives of $f(b)$ with respect to b :

(a)

$$\frac{\partial f(b)}{\partial b},$$

(b)

$$\frac{\partial^2 f(b)}{\partial b \partial b^\top},$$

Solution

The function $f(b)$ maps $b \in \mathbb{R}^{(p+1)}$ to the scalar $f(b) \in \mathbb{R}$. To see that $f(b)$ is a scalar, observe that

$$\begin{aligned}
 f(b) &= (y - Xb)^\top (y - Xb) \\
 &= (y^\top - (Xb)^\top)(y - Xb) \\
 &= y^\top y - \underbrace{y^\top Xb}_{(i)} - \underbrace{(Xb)^\top y}_{(ii)} + \underbrace{(Xb)^\top Xb}_{(iii)}
 \end{aligned}$$

where

$$(i) \quad y^\top Xb = (X^\top y)^\top b$$

$$\begin{aligned}
 (ii) \quad \underbrace{(Xb)^\top y}_{(1 \times 1)} &= \underbrace{((Xb)^\top y)^\top}_{(1 \times 1)} \quad \left| \text{[Scalars are always symmetric]} \right. \\
 &= y^\top (Xb) \\
 &= (y^\top X)b \\
 &= (X^\top y)^\top b
 \end{aligned}$$

$$\begin{aligned}
 (iii) \quad (Xb)^\top Xb &= (Xb)^\top Xb \\
 &= b^\top X^\top Xb.
 \end{aligned}$$

Thus, we can write $f(b)$ as

$$f(b) = y^\top y - 2(X^\top y)^\top b + b^\top X^\top X b$$

(a) Derivative rule (i) of our script:

$$\frac{\partial a^\top b}{\partial b} = a \quad \text{here:} \quad \frac{\partial (X^\top y)^\top b}{\partial b} = X^\top y$$

Derivative rule (ii) of our script:

$$\frac{\partial b^\top A b}{\partial b} = (A + A^\top) b \quad \underbrace{= 2Ab}_{\text{if } A \text{ symmetric}}$$

Here:

$$\frac{\partial b^\top (X^\top X) b}{\partial b} = ((X^\top X) + (X^\top X)^\top) b = \underbrace{((X^\top X) + (X^\top X))}_{\text{since } X^\top X \text{ is symmetric}} b = 2X^\top X b$$

Thus,

$$\begin{aligned} \frac{\partial f(b)}{\partial b} &= \overbrace{\frac{\partial y^\top y}{\partial b}}^{=0} - 2 \overbrace{\frac{\partial (X^\top y)^\top b}{\partial b}}^{=X^\top y} + \overbrace{\frac{\partial b^\top X^\top X b}{\partial b}}^{=2X^\top X b} \\ &= 2X^\top X b - 2X^\top y \end{aligned}$$

(b) Derivative rule (iii) of our script (Hessian of a sandwich form function):

$$\frac{\partial^2 (b^\top A b)}{\partial b \partial b^\top} = A + A^\top \quad \underbrace{= 2A}_{\text{if } A \text{ symmetric}}$$

Here:

$$\frac{\partial^2 b^\top X^\top X b}{\partial b \partial b^\top} = X^\top X + (X^\top X)^\top = 2X^\top X \quad \text{since } X^\top X \text{ is symmetric}$$

Thus

$$\begin{aligned} \frac{\partial^2 f(b)}{\partial b \partial b^\top} &= \frac{\partial^2}{\partial b \partial b^\top} (y^\top y - 2(X^\top y)^\top b) + \underbrace{\frac{\partial^2 b^\top X^\top X b}{\partial b \partial b^\top}}_{=2X^\top X} \\ &= \frac{\partial}{\partial b^\top} \left(\frac{\partial}{\partial b} (y^\top y - 2(X^\top y)^\top b) \right) + 2X^\top X \\ &= \frac{\partial}{\partial b^\top} (-2X^\top y) + 2X^\top X \\ &= 2X^\top X \end{aligned}$$

Note:

$$\frac{\partial^2 f(b)}{\partial b \partial b^\top} = 2X^\top X > 0$$

if X has full column rank $\text{rank}(X) = p + 1$.